MAT 3103: Computational Statistics and Probability
Semester: Fall 2019-20
Chapter 3[Random Variable and Moments]

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Random variable

3.1 Random Variable: A Random Variable is a function, which assigns unique numerical values to all possible outcomes of arandom experiment under fixed conditions. A random variable is not a variable but rather a function that maps events to numbers.A random variable, usually written *X*, is a variable whose possible values are numerical outcomes of a random phenomenon.

We often use letters like *X*, *Y* and *Z* to denote a random variable. Here is an example

Example

Suppose that a coin is tossed two times and the sequence of heads and tails is noted. The sample space for this experiment evaluates to: S={TT,TH,HT,HH}. Now let the random variable *X* be the number of heads in two coin tosses. *X* assigns each outcome in S a number from the set S*x*={0, 1, 2}. The table below lists the eight outcomes of S and the corresponding values of *X*.

Outcome	TT	TH	HT	HH
X	0	1	1	2

X is then a random variable taking on values in the set S_X= {0, 1, 2}.

Which takes the values 0,1,2 with probabilities –

$P(X = 0)$ =P (no head)= $\frac{1}{4}$

$P(X = 1) = \frac{1}{2}$

$P(X = 2) = \frac{1}{4}$

3.2 Types of random variables

There are two types of random variables

- **3.2.1** ***Discrete random variable***
- **3.2.2** ***Continuous random variable***

3.2.1 Discrete random variable: A discrete random variable can take on only specific, isolated numerical values such as 0,1,2,3,4, ... , like the outcome of a roll of a die, or the number of dollars in a randomly chosen bank account. Discrete random variables that can take on only finitely many values (like the outcome of a roll of a die) are called finite random variables. Discrete random variables that can take on an unlimited number of values (like the number of stars estimated to be in the universe) are infinite discrete random variables.

3.2.2 Continuous random variable: A continuous random variable, on the other hand, can take on any values within a continuous range or an interval, like the temperature in Dhaka city, or the height of student of the class. It takes an infinite number of possible values. Continuous random variables are usually measurements. A continuous random variable is not defined at specific values.

3.3 Probability function

A probability function for a discrete random variable is called ***a discrete probability function***.

Let X be a discrete random variable with possible values $x_1, x_2 \ldots\ldots$ Let the corresponding probabilities are $p(x_1)$, $p(x_2) \ldots\ldots\ldots$ If the functions $p(x_i)$ satisfies the conditions,

$$(i)\boldsymbol{p(x)} \geq \mathbf{0}$$
$$(ii)\sum_{\mathbf{0}}^{\infty} \boldsymbol{p(x)} = \mathbf{1}$$

Then $p(x)$ is called a ***probability mass function*** and X has a discrete probability function.

A probability function for a continuous random variable is called ***a continuous probability function***.

Let X be a continuous random variable with possible values $x_1, x_2 \ldots\ldots\ldots$ Let the corresponding probabilities are $f(x_1), f(x_2) \ldots\ldots\ldots$ If the functions $f(x_i)$ satisfies the conditions,

$$(i)\boldsymbol{f(x)} \geq \mathbf{0}$$
$$(ii)\int_{-\infty}^{\infty} \boldsymbol{f(x)dx} = \mathbf{1}$$

Then $f(x)$ is called a ***probability density function*** and x has a continuous probability function.

As for example, $f(x) = 2x$; $0 \leq x \leq 1$. Here, $\{x, f(x)\}$ is called probability density function.

If X and Y are ***two discrete random variables*** and $p(x, y)$ is a function of X and Y such that-

(i) $0 \leq p(x,y) \leq 1$, (ii) $\sum_{x=0}^{\infty} \sum_{y=0}^{\infty} p(x,y) = 1$

then, $p(x, y)$ is called **joint probability function**.

Marginal probability function of X and Y are respectively given by

$g(x)=\sum_j P(X = x_i; Y = y_j)$

$h(y)=\sum_i P(X = x_i; Y = y_j)$

If X and Y are ***two continuous random variables*** and $f(x, y)$ is a function of X and Y such that-

(i) $0 \leq f(x,y) \leq 1$, (*ii*) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dxdy = 1$

then, $f(x,y)$ is called **joint probability density function**.

Marginal probability function of X and Y are respectively given by

$$g(x)=\int_y f(x,y)dy$$

$$h(y)=\int_x f(x,y)dx$$

If $f(xy) = g(x).h(y)$, then X and Y are independent.

3.4 Mean and variance of random variable

The average value of a random variable is called the expected value of the random variable. Expectation is very important for decision making.

Let X be the random variable either discrete or continuous with probability function $p(x)$ and $f(x)$ respectively. Then expectation of x is defined as

$$E(x) = \sum_0^\infty \boldsymbol{xp}(\boldsymbol{x}) ; \quad \text{if } x \text{ is discrete}$$

$$=\int_{-\infty}^{\infty} \boldsymbol{xf}(\boldsymbol{x})\boldsymbol{dx}; \quad \text{if } x \text{ is continuous}$$

Variance of x is given by-

$$V(x) = E(x^2) - [E(x)]^2$$

Where, $E(x^2) = \sum_{x=0}^{\infty} \boldsymbol{x^2 p}(\boldsymbol{x})$, if x is discrete

$= \int_{-\infty}^{\infty} \boldsymbol{x^2 f}(\boldsymbol{x})dx$, if x is continuous.

3.5 Properties of Mean and Variance:

1. If A is any constant and X is a random variable, then
   - $E(A) = A$,
     $V(A) = 0$.
   - $E(AX) = A\,E(X)$,
     $V(AX) = A^2 V(X)$.

2. If A and B are any two constants and X is a random variable, then
   $E(AX \pm B) = AE(X) \pm B$,
   $V(AX \pm B) = A^2 V(X)$.

3. If A and B are any two constants and X and Y are two random variables, then
   $E(AX \pm BY) = AE(X) \pm BE(Y)$,

$V(AX \pm BY) = A^2 V(X) + B^2 V(Y) \pm 2AB\,Cov(XY)$ [if X and Y are not independent]

$V(AX \pm BY) = A^2 V(X) + B^2 V(Y)$ [if X and Y are independent, then $Cov(XY)=0$]

4. If X and Y are two independent random variables, then
$E(XY) = E(X)\,E(Y)$

Covariance of X and Y:

Covariance of X and Y is given by-
$Cov(XY) = E(XY) - E(X)E(Y)$
Where,

$E(XY) = \sum_{x=0}^{\infty} \sum_{y=0}^{\infty} xy\, p(x,y)$, if x and y are discrete

$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x,y)\, dxdy$, if x and y are continuous

If X and Y are independent random variables, then using the 4th property $Cov(XY) = 0$.

Example.1:

Given, the probability function of x as follows:

x	0	1	2	3
p(x)	0.2	0.3	0.2	0.3

Calculate (i) P($x \leq 2$), (ii) P($x>1$), (iii) P($1< x <3$), (iv) $E(3x-2)$, (v) $V(3X+4)$.

Solution:

(i) $P(x \leq 2) = P(x=0) + P(x=1) + P(x=2)$ or, $P(x \leq 2) = 1 - P(x=3)$

$= 0.2+0.3+0.2$ $= 1 - 0.3$

$= 0.7$ $= 0.7$

(ii) $P(x>1) = P(x=2) + P(x=3)$
$= 0.2+0.3$
$= 0.5$

(iii) $P(1< x <3) = P(x=2) = 0.2$

(iv) $E(3x-2) = 3E(x) - 2 = (3\times1.6) - 2 = 2.8$

where, $E(x) = \sum_{X=0}^{3} xf(x) = (0\times0.2)+(1\times0.3)+(2\times0.2)+(3\times0.3) = 1.6$

(v) $V(3x+4) = 3^2V(x)+0 = 9V(x) = 9\times1.24 = 11.16$

where, $V(x) = E(x^2) - [E(x)]^2 = 3.8 - (1.6)^2 = 1.24$

Here, $E(x) = 1.6$

and $E(x^2) = \sum_{X=0}^{\infty} x^2 f(x) = (0^2 \times0.2)+(1^2 \times0.3)+ (2^2 \times0.2)+ (3^2 \times0.3) = 3.8$

Example.2:

Given, the probability density function of X as follows:

$f(x) = \frac{x}{2}$; $0<x<2$

Find (i) $P(x<1)$, (ii) $P(x>0.5)$, (iii) $P(1< x <2)$, (iv) $E(2x+3)$, (v) $V(3x-2)$.

Solution: (i) $P(x<1) = \int_0^1 f(x)\, dx = \int_0^1 \frac{x}{2}\, dx = [\frac{X^2}{4}]_0^1 = \frac{1}{4}$

(i) $P(x>0.5) = \int_{0.5}^2 f(x)\, dx = \int_{0.5}^2 \frac{X}{2}\, dx = [\frac{X^2}{4}]_{0.5}^2 = [\frac{2^2}{4} - \frac{0.5^2}{4}] = 0.9375$

(ii) $P(1< x <2) = \int_1^2 \frac{X}{2}\, dx = [\frac{X^2}{4}]_1^2 = [\frac{2^2}{4} - \frac{1^2}{4}] = \frac{3}{4}$

(iii) $E(2x+3) = 2E(x) + 3 = (2\times \frac{4}{3}) + 3 = 5$

where, $E(x) = \int_X x\, f(x)\, dx = \int_0^2 x \frac{X}{2}\, dx = \int_0^2 \frac{X^2}{2}\, dx = [\frac{X^3}{6}]_0^2 = \frac{4}{3}$

(iv) $V(3x-2) = 9V(x) = (9\times \frac{2}{9}) = 2$

where, $V(x) = E(x^2) - [E(x)]^2 = 2-(\frac{4}{3})^2 = \frac{2}{9}$

here, $E(x^2) = \int_0^2 x^2 \frac{X}{2}\, dX = \int_0^1 \frac{X^3}{2}\, dx = [\frac{X^4}{8}]_0^2 = 2$

Example.3:

The joint probability density function of X and Y is as follows:

$f(x, y) = 2x$, $0<x<1$, $0<y<1$

Calculate- (i) marginal distribution of X and Y, (ii) $P(X>\frac{1}{2})$, (iii) $P(\frac{1}{3}<X<\frac{1}{2})$, (iv) $P(Y>\frac{1}{2})$, (v) $E(3X - 2)$, (vi) $V(2X - 3)$, (vii) $E(X - 2Y)$, (viii) $V(2X - Y)$.

Solution: We have, Marginal distribution of X is,

$g(x) = \int_y f(x,y)\, dy = \int_0^1 2x dy = 2x \int_0^1 1\, dy = 2x\, [y]_0^1 = 2x(1-0) = 2x.$

Marginal distribution of Y is,

$h(y) = \int_X f(x,y)\, dx = \int_0^1 2x\, dx = [\frac{2X^2}{2}]_0^1 = 1$

Since, g(x) h(y) = $2x \times 1 = 2x = f(x,y)$

So, X&Y are independent random variables; i.e.$Cov\,(X\text{Y}) = 0$.

(i) $\text{P}\,(x>\frac{1}{2}) = \int_{\frac{1}{2}}^{1} g(x)\,dx = \int_{\frac{1}{2}}^{1} 2x\,dx = [\frac{2X^2}{2}]_{\frac{1}{2}}^{1} = 1-\frac{1}{4} = \frac{3}{4}$

(ii) $\text{P}\,(\frac{1}{3}<x<\frac{1}{2}) = \int_{\frac{1}{3}}^{\frac{1}{2}} 2x\,dx = [\frac{2X^2}{2}]_{\frac{1}{3}}^{\frac{1}{2}} = \frac{1}{4}-\frac{1}{9} = \frac{5}{36}$

(iii) $E(3x\text{-}2) = 3E(x) - 2 = (2\times\frac{2}{3}) - 2$

where, $E(x) = \int_X x\,g(x)\,dx = \int_0^1 x\,2x\,dx = \int_0^1 2x^2\,dx = [\frac{2X^3}{3}]_0^1 = \frac{2}{3}$

(iv) $V(2x\text{-}3) = 2^2V(x) - 0 = 4V(x) = 4\times\frac{1}{18} = \frac{2}{9}$

where, $V(x) = E(x^2) - [E(x)]^2 = \frac{1}{2} - (\frac{2}{3})^2 = \frac{1}{18}$

here, $E(x^2) = \int_X x^2 g(x)\,dx = \int_0^1 x^2\,2x\,dx = \int_0^1 2x^3\,dx = [\frac{2X^4}{4}]_0^1 = \frac{1}{2}$

(v) $E(x-2y) = E(x) - 2E(\text{y}) = \frac{2}{3} - (2\times\frac{1}{2}) = -\frac{1}{3}$

where, $E(y) = \int_{\text{y}} \text{y}\,h(\text{y})\,d\text{y} = \int_0^1 \text{y}\,d\text{y} = [\frac{y^2}{2}]_0^1 = \frac{1}{2}$

(vi) $V(2x\text{-}y) = 2^2V(x) + V(y) - 2.2.1\,\text{Cov}(xy) = 4V(x) + V(y)$ [as $\text{Cov}(xy) = 0$]

$V(y) = E(y^2) - [E(y)]^2 = \frac{1}{3} - (\frac{1}{2})^2 = \frac{1}{12}$

$E(y^2) = \int_{\text{y}} y^2\,h(y)\,dy = \int_0^1 y^2\,dy = [\frac{y^3}{3}]_0^1 = \frac{1}{3}$

$\therefore V(2x\text{-}y) = (4\times\frac{1}{18}) + \frac{1}{12} = \frac{11}{36}$

E*X*ample.4:

The joint probability function of two random variables X and Y is shown below:

X/Y	0	1	2	g(x)
0	0.1	0.2		0.3
1	0.1	0.2	0.1	0.4
2		0.1	0.1	0.2
3	0.1			0.1
h(y)	0.3	0.5	0.2	1

Calculate- (i) P(X<2) , (ii) P(0<X<3) , (iii) P(Y≥1) , (iv) E(2X-Y).

Solution:

(i) $P(x<2) = 0.3 + 0.4 = 0.7$

(ii) $P(0<x<3) = 0.4 + 0.2 = 0.6$

(iii) $P(y\geq 1) = 0.5 + 0.2 = 0.7$

(iv) $E(2x-y) = 2E(x) - E(y) = (2\times 1.1) - 0.9 = 1.3$

Where, $E(x) = \sum_X x\ g(x) = (0\times 0.3) + (1\times 0.4) + (2\times 0.2) + (3\times 0.1) = 1.1$

and $E(y) = \sum_y y\ h(y) = (0\times 0.3) + (1\times 0.5) + (2\times 0.2) = 0.9$

3.6 Sample questions related to this chapter

1. The probability function of a random variable of X as follows:

X	0	1	2	3
p(x)	0.3	0.2	0.3	0.2

Calculate (i) $P(X\leq 3)$, (ii) $P(X>2)$, (iii) $P(1< X <3)$, (iv) $E(3X-1)$, (v) $V(2X+4)$.

2. The joint probability function of two random variables X and Y is shown below:

X/Y	0	1	2	3	g(x)
1	0.2	0.1	0.1	0.0	0.4
2	0.1	0.0	0.1	0.1	0.3
3	0.0	0.2	0.0	0.1	0.3
h(y)	0.3	0.3	0.2	0.2	1

Calculate- (i) $P(X<1)$, (ii) $P(0<X<2)$, (iii) $P(Y\geq 2)$, (iv) $E(X-2Y)$.

3. Given, the probability density function of X as follows:
$f(x) = 2x$; $0< x <1$

Find (i) $P(X<1)$, (ii) $P(X>0.5)$, (iii) $E(X+2)$, (iv) $V(2X-1)$.

4. The joint probability density function of X and Y is as follows:

   $f(x, y) = 4xy, \quad 0<x<1, \ 0<y<1$

   Calculate- (i) P $(0<X< 0.5)$, (ii) P $(Y < 0.75)$, (iii) $E(2X - 3)$, (iv) $V(3X - 4)$, (v) $E(X - 3Y)$, (vi) $V(2X - Y)$.

Moments of random variable and shape characteristics of distribution

The moments of a random variable (or of its distribution) are mean deviation of powers or related functions of the random variable.

Let Xbe a random variable with probability function $f(X)$.

The r[th] moment of X is $E(X - A)^r$. Then

$$\mu_{[r]} = E(x - A)^r = \sum_{x=0}^{\infty}(x - A)^r f(x); \text{ if } X \text{ is discrete}$$
$$= \int_{-\infty}^{\infty}(x - A)^r f(x)dx; \text{ if } X \text{ is continuous}$$

where r =0, 1, 2, 3, 4, … a finite number, say k. $\mu_{[r]}$ is called general moment. Deviation from any arbitrary value (A). Where A is a constant.

We can also define, the r[th] order mean deviation about zero (0).

$$\mu_r^{'} = E(x - 0)^r = \sum_{x=0}^{\infty}(x - 0)^r f(x); \text{ if } X \text{ is discrete}$$
$$= \int_{-\infty}^{\infty}(x - 0)^r f(x)dx; \text{ if } X \text{ is continuous}$$

$\mu_r^{'}$ is called raw moment.

The r[th] order mean deviation about mean is defined as

$$\mu_r = E(x - \mu)^r = \sum_{x=0}^{\infty}(x - \mu)^r f(x); \text{ if } X \text{ is discrete}$$
$$= \int_{-\infty}^{\infty}(x - \mu)^r f(x)dx; \text{ if } X \text{ is continuous}$$

μ_r is called the central moment. If A=0, $\mu_{[r]} = \mu_r^{'}$and if A=μ,$\mu_r^{'} = \mu_r$.

When r = 0, the moment of zero order is always 1.

When r = 1, then

$$\mu_{[1]} = E(x - A)^1 = \int_{-\infty}^{\infty}(x - A)^1 f(x)dx = \int_{-\infty}^{\infty} xf(x)dx - A\int_{-\infty}^{\infty} f(x)dx = \mu - A$$

Which is mean deviation of X about any arbitrary value A. If A = 0

$\mu_1^{'} = E(X) = \int_{-\infty}^{\infty} xf(x)dx$ = mean = μ [first raw moment is actually mean]

Again,

$\mu_1 = E(x-\mu)^r = \int_{-\infty}^{\infty}(x-\mu)^r f(x)dx = \int_{-\infty}^{\infty} xf(x)dx - \mu\int_{-\infty}^{\infty} f(x)dx$ = μ - μ= 0

First central moment is always 0 for any dfistribution.

Let *r* = 2,

$$\mu_2 = E(x-\mu)^2 = E(x)^2 + E(\mu)^2 - 2\mu E(x)$$
$$= \mu_2^{'} + {\mu_1^{'}}^2 - 2{\mu_1^{'}}^2$$
$$= \mu_2^{'} - {\mu_1^{'}}^2$$

So, $\mu_2$is actually variance,

In the same manners,

If *r* = 3,

$$\mu_3 = E(x-\mu)^3$$
$$= \mu_3^{'} - 3\mu_2^{'}\mu_1^{'} + 2{\mu_1^{'}}^3$$

If *r* = 4,

$$\mu_4 = E(x-\mu)^4$$
$$= \mu_4^{'} - 4\mu_3^{'}\mu_1^{'} + 6\mu_2^{'}{\mu_1^{'}}^2 - 3{\mu_1^{'}}^4$$

The central moments are used to study the shape characteristic of any distribution.

Different frequency distribution or probability may take different shapes when these are presented by frequency curve. We can explain the shapes of curve by skewness and kurtosis.

Skewness: The asymmetric property of frequency distribution is known as skewness.

Coefficient of skewness(in term of moments),

Coefficient of skewness, $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$, $\gamma_1 = \sqrt{\beta_1}$,

if, $\gamma_1 = 0, \mu_3$ =0; the distribution is symmetric
$\gamma_1 > 0, \mu_3$ >0; the distribution is right skewed
$\gamma_1 < 0, \mu_3$ <0; the distribution is left skewed

Kurtosis: Height characteristics of frequency curve is known as kurtosis.

Coefficient of kurtosis

$$\text{Coefficient of kurtosis}, \beta_2 = \frac{\mu_4}{\mu_2^2} \ ,$$

if, β_2 =3; (the distribution is mesokurtic)
β_2 >3; (the distribution is leptokurtic)
β_2 <3; (the distribution is platykurtic)

